

5 th Semester		
ISCED Code	Course Code	Course Title
0613	CSE-3529	System Analysis and Design
Credit Hours: 3	Contact Hours: 3	Type: Core, Engineering
Prerequisite: N/A		
Co-requisite: N/A		

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Course Assessments	CIE:	Attendance	10 Marks
	Continuous Internal Evaluation	Class test/ Assignment/ Quizzes	10 Marks
		Mid-term	30 Marks
	SEE: Semester End Examination		50 Marks
	Total		100 Marks

Course Rationale:

The main goal of this System Analysis and Design course is to provide system analysis and design techniques to the students to carry out advance software engineering and software development courses throughout the career. Coursework consists of discussion questions, group presentations, mid-term and final written exam with essay-style questions. By those means it is expected that participants will also improve their verbal and written communication skills, as well as their presentation skills.

Objectives:

1.	Upon completion of the course, students will be able to:
1.	To understand the traditional systems/software development methodologies and design techniques: entity-relationship diagrams, data flow diagrams, and data dictionaries.
2.	To be able to use prototyping and different tools like CASE in the analysis and design phases of systems development.
3.	To design project planning: defining the scope, purpose and activities of a project; setting up a project management.
4.	To understand how Object-Oriented methodology techniques work

Course Learning Outcomes (CLOs):

Upon successful completion of this course, students will be able to:

#	CLO Description	Weightage
1	Understand the fundamentals of system development process	20%
2	Analyze different kinds of system models testing mechanism, implementation and maintenance techniques for assuring system quality and system security	30%
3	Apply different kinds of system models with user requirements and feasibility	20%
4	Formulate the structured design tools (DFD, ERD) and analyze different design techniques for building any system	30%

Mapping of CO-PO-WK-WP-WA:

#	COs	POs	DL	K P	E P	E A	Teaching Learning Strategy (TLS)	Assessment Strategy (AS)
CLO 1	Understand the fundamentals of system development process	PLO 1	Cognitive/Underst and	K3 , K4	-	-	Lecture, Class discussion , Assignme nt, Note	Exam, Quiz, Assignment

#	COs	POs	DL	K P	E P	E A	Teaching Learning Strategy (TLS)	Assessment Strategy (AS)
CLO 2	Analyze different kinds of system models testing mechanism, implementati on and maintenance techniques for assuring system quality and system security	PLO 2	Cognitive/Analyze	K4	-	-	Lecture, Class discussion , Assignme nt, Note	Exam, Quiz, Assignment
CLO 3	Apply different kinds of system models with user requirements and feasibility	PLO 2	Cognitive/Apply	K3	-	-	Lecture, Class discussion , Assignme nt, Lab work, Note, Complex Problem Solving	Exam, Quiz, Assignment,
CLO 4	Formulate the structured design tools (DFD, ERD) and analyze different design techniques for building any system	PLO 2	Cognitive/Apply	K4			Compile the system design document	System design documentati on

#	COs	POs	DL	K P	E P	E A	Teaching Learning Strategy (TLS)	Assessment Strategy (AS)
Note: DL: Domain/level of learning taxonomy, KP: Knowledge Profile, EP: Attribute of Complex Engineering Problems, EA: Attribute of Complex Engineering Activities Learning Domains (C: Cognitive, A: Affective, P: Psychomotor)								

Course Content:

No	Content	Duration	CLOs
1.	Concepts of system and its environment: Information, Types of information, Quality of information, System, Types of systems, Components of system, Source of information.	02 weeks	CLO1
2.	Information gathering: Information gathering strategy, Information searching methods, Interviewing technique, System development methodologies and life cycle.	02 weeks	CLO3
3.	Feasibility study & Cost/Benefit analysis: Feasibility considerations, steps in feasibility analysis, feasibility report, Cost and Benefit categories, procedure for cost and benefit determination, classification of cost and benefit, cost and benefit evaluation methods.	02 weeks	CLO3
	Final: Part A (20 Marks)		
4.	Tools of analysis and design: Data Flow Diagram (DFD), DFD symbols, Constructing DFD; Data Dictionary; Decision Tree, Structured English, Decision Tables.	02 weeks	CLO4
5.	System Design and Construction: The process of design, System design phases, Design methodologies; Structured design; Form-Driven methodology; Input design, Output design, File and database design	02 weeks	CLO4

	Final: Part B (30 Marks)		
6.	Testing and Quality Assurance: Testing, Types of system tests; White-Box testing; Black-box testing; Quality factors specifications.	01 week	CLO2
7.	Implementation and Maintenance: Types of implementations, Documenting the system, Training and supporting users, Factor models of implementation success; The process of maintaining information system, Types of maintenance, Cost of maintenance, Reducing maintenance cost.	02 weeks	CLO2
8.	Hardware/ Software selection, control and security: Phases in selection, Criteria for software selection, Hardware selection, Financial considerations in selection; Security definitions, Threats to system security, Control measures, system failures and recovery.	02 weeks	CLO2

Marks Distribution:

Course Assessment Pattern (Theory courses):

Bloom's Category		Evaluations out of 100 marks			
		CIE (50 marks)			SEE (50 marks)
Cognitive learning	Affective learning	Attendance Marks (10)	Assignment/ Class Test (10)	Mid-term (30)	Written Exam (50)
Remember					5
Understand			5	5	15
Apply			5	10	
Analyze				15	30
Evaluation					
Create					
	Responding	10			
Total allocated marks		10	10	30	50

Grading Policy: As per IIUC grading policy

Learning Materials:

Text Books:

#	Name of Authors	Title of Book	Edition	Publisher's Name	Year	ISBN
1	Jeffrey Whitten Lonnie D. Bentley"	Systems Analysis and Design Methods	7th Edition	McGraw-Hill Education"	2007	0073052337, 9780073052335
2	Kenneth E. Kendall Julie E. Kendall	Systems Analysis and Design	8th Edition	Prentice Hall	2010	13: 978-0136089162

Reference Books:

#	Name of Authors	Title of Book	Edition	Publisher's Name	Year	ISBN
1	Alan Dennis Barbara Haley Wixom Roberta Marie Roth	Systems Analysis and Design	5th	Wiley	2012	1118057627, 9781118057629
2	Akhilesh Bajaj	Systems Analysis and Design for Advanced Modeling Methods: Best Practices	1st Edition	IGI Global	2009	978-1-60566-344-9
3	Alan Dennis Barbara Haley Wixom	Systems Analysis and Design	11th Edition	Edition "John Wiley & Sons Inc."	2008	13: 978-0137689958"

	Roberta M. Roth					
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Lecture Plan

Week	Main Topic	Topics / Contents	CLO	Assessment / Remarks
Week 1	Concepts of System and Its Environment	Information; Types of Information; Quality of Information; System and Types of Systems	CLO1	Lecture
Week 2	Concepts of System and Its Environment	Components of System; Sources of Information; System Environment; Review of Concepts	CLO1	Lecture + Class Discussion
Week 3	Information Gathering	Information Gathering Strategy; Information Searching Methods; Interviewing Techniques	CLO3	Lecture + Practical Examples
Week 4	Information Gathering	System Development Methodologies; System Development Life Cycle (SDLC); Comparison of Methodologies	CLO3	Lecture + Class Discussion
Week 5	Feasibility Study & Cost/Benefit Analysis	Feasibility Considerations; Steps in Feasibility Analysis; Feasibility Report	CLO3	Lecture + Case Study
Week 6	Feasibility Study & Cost/Benefit Analysis	Cost and Benefit Categories; Cost/Benefit Determination; Classification; Evaluation Methods	CLO3	Lecture + Problem Solving
	MID-TERM EXAMINATION	Topics 1–3	CLO1, CLO3	Mid-Term Examination
Week 7	Tools of Analysis and Design	Data Flow Diagram (DFD); DFD Symbols; Rules for Constructing DFD	CLO4	Lecture + DFD Practice

Week 8	Tools of Analysis and Design	Data Dictionary; Decision Tree; Structured English; Decision Tables	CLO4	Lecture + Problem Solving
Week 9	System Design and Construction	Process of Design; System Design Phases; Design Methodologies; Structured Design	CLO4	Lecture
Week 10	System Design and Construction	Form-Driven Methodology; Input Design; Output Design; File and Database Design	CLO4	Lecture + Design Examples
Week 11	Testing and Quality Assurance	Testing Concepts; Types of System Tests; White-Box Testing; Black-Box Testing; Quality Factors	CLO2	Lecture + Examples
Week 12	Implementation and Maintenance	Types of Implementation; System Documentation; Training and Supporting Users; Factors Affecting Implementation Success	CLO2	Lecture + Case Study
Week 13	Implementation and Maintenance	Information System Maintenance Process; Types of Maintenance; Cost of Maintenance; Reducing Maintenance Cost	CLO2	Lecture + Discussion
Week 14	Hardware/Software Selection, Control and Security	Selection Phases; Software Selection Criteria; Hardware Selection; Financial Considerations	CLO2	Lecture + Case Study
Week 15	Hardware/Software Selection, Control and Security	Security Definitions; Security Threats; Control Measures; System Failures and Recovery	CLO2	Lecture + Review
—	FINAL EXAMINATION	Topics 4–8	CLO2, CLO4	Final Examination